

Hsin-Jung Hsieh

hsinjunghsieh@ucsb.edu • +1(805)8967134

EDUCATION

University of California, Santa Barbara

M.S. in Electrical and Computer Engineering (CE emphasis, GPA: 3.86/4.0)

CA, USA

Degree expected June 2026

- Focus: Machine Learning, Deep Learning, and AI in Design Automation
- Research: GANs for adversarial attacks on LLMs; developing self-learning agent architectures for adaptive reasoning and domain-specific knowledge transfer; robustness of large-scale generative models

National Cheng Kung University

B.S. in Computer Science and Information Engineering

Tainan, Taiwan

June 2024

- Research Assistant, Ubiquitous Sensing and Cloud Computing Lab

PROJECTS

Imeiskin: AI-Powered Dermatological Self-Diagnosis System, Lab Research | NCKU, Taiwan

- Designed and implemented a novel AI-based healthcare platform enabling non-clinical users to perform preliminary self-assessment for dermatological conditions (skin, acne, nails, tongue) through image uploads.
- Trained and evaluated state-of-the-art CNN architectures (DenseNet121, ResNet101, custom CNN), achieving 84.7% classification accuracy across diverse datasets curated from open-access medical imagery. The models generated confidence scores per prediction; at $\geq 90\%$ confidence, diagnoses correlated with 94% accuracy, supporting reliable clinical decision making.
- Developed a cross-platform ecosystem including responsive web apps, Android mobile interfaces, and a Line chatbot, empowering real-time interaction, diagnosis history tracking, and progressive monitoring.

Web-Integrated HR & Payroll Management System, Software Engineering Intern Project

- Spearheaded the development of a robust HRMS solution integrating real-time card-swipe, leave, overtime, and payroll modules using VB and SQL.
- Designed advanced back-end logic for automated calculation of allowances and user-configurable payroll deductions, reducing human errors, saving processing time and improving flexibility.
- Implemented reporting capabilities (payroll summaries, compliance-ready outputs) and interactive UI dashboards with administrative override controls, making the system scalable for enterprise deployment.

Intelligent Chatbot Audio System with Emotion Recognition, Raspberry Pi Design Side Project

- Developed a Python-based conversational AI assistant integrating NLU, voice interaction, and emotion recognition modules.
- Designed a modular pipeline for dialogue management and context-aware responses, simulating real-time LLM-style conversational flow.
- Deployed the system on Raspberry Pi with multimodal outputs (OLED expressive animations + voice), demonstrating applications in human-computer interaction and affective computing.
- Extended to incorporate reinforcement-style self-learning for adaptive dialogue behaviors, forming the foundation for ongoing research on self-learning AI agents.

INTERNSHIP EXPERIENCE

Finecode Technology Co., Ltd.

New Taipei, Taiwan

Software Engineering Intern

Summer 2023-2025

- Engineered two systems (HRMS and Construction Scheduler), streamlining operations across multi-branch organizations.
- Deployed optimized database schemas, implemented secure login/user access control, and led backend logic development.
- Awarded "Intern of the Month" twice for outstanding contributions to project delivery and innovation.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Dart, Verilog, Linux, VB, SQL

ML/DL: PyTorch, TensorFlow, LangChain, AutoGen, vLLM, HuggingFace Accelerate & Transformers, Scikit-learn

AI Domains: Computer Vision, NLP, LLMs, Adversarial ML, GANs, AI for Healthcare

Tools: Git, Linux, MATLAB, Android Studio, Cadence, ModelSim

Other: Web Development, Database Systems, VLSI Design, FPGA Design

ACTIVITIES

Google Developer Student Club (ML Group)

Jan 2022 – Sep 2022.

- Participated in hands-on workshops covering data preprocessing, model training, hyperparameter tuning, and evaluation using Python and ML libraries (Scikit-learn, TensorFlow, PyTorch).
- Collaborated with peers on small-scale ML projects such as text classification and image recognition, building foundational skills in supervised/unsupervised learning and neural networks.
- Gained exposure to applied AI workflows that supported later research in GANs, adversarial attacks, and LLMs.